



SehatSaathi AI

Your Personal AI Healthcare Companion

Empowering Pakistan with Intelligent, Accessible Healthcare



Hackathon 2026 Submission | Team Members: **Maryam Tahir**

The Problem

- ⚠️ 73% of rural Pakistan lacks specialist access
- ⚠️ Average diagnostic wait: 3-7 days
- ⚠️ Language barrier in healthcare (Urdu speakers)
- ⚠️ Prescription misinterpretation risk

Our Solution

- ✅ AI diagnostics accessible 24/7 from any device
- ✅ Results in under 5 seconds
- ✅ Full Urdu & English bilingual support
- ✅ Smart prescription OCR with medicine ordering

SehatSaathi AI — 8 Healthcare Modules

AI Diagnostics

MRI, X-Ray & Skin Analysis
ONNX CNN Models • ~99% accuracy

Symptom Checker

132 symptoms → 41 diseases
ONNX SVC ML Model

AI Chat (EN/UR)

Groq LLaMA 3.3-70b
Bilingual Urdu + English

Prescription OCR

Tesseract.js — offline
Photo → Medicine table → Cart

Pharmacy Store

15,000+ real medicines
COD checkout, search & filter

Hospital Locator

OpenStreetMap via Leaflet
Filter by city & specialty

Diet Planner

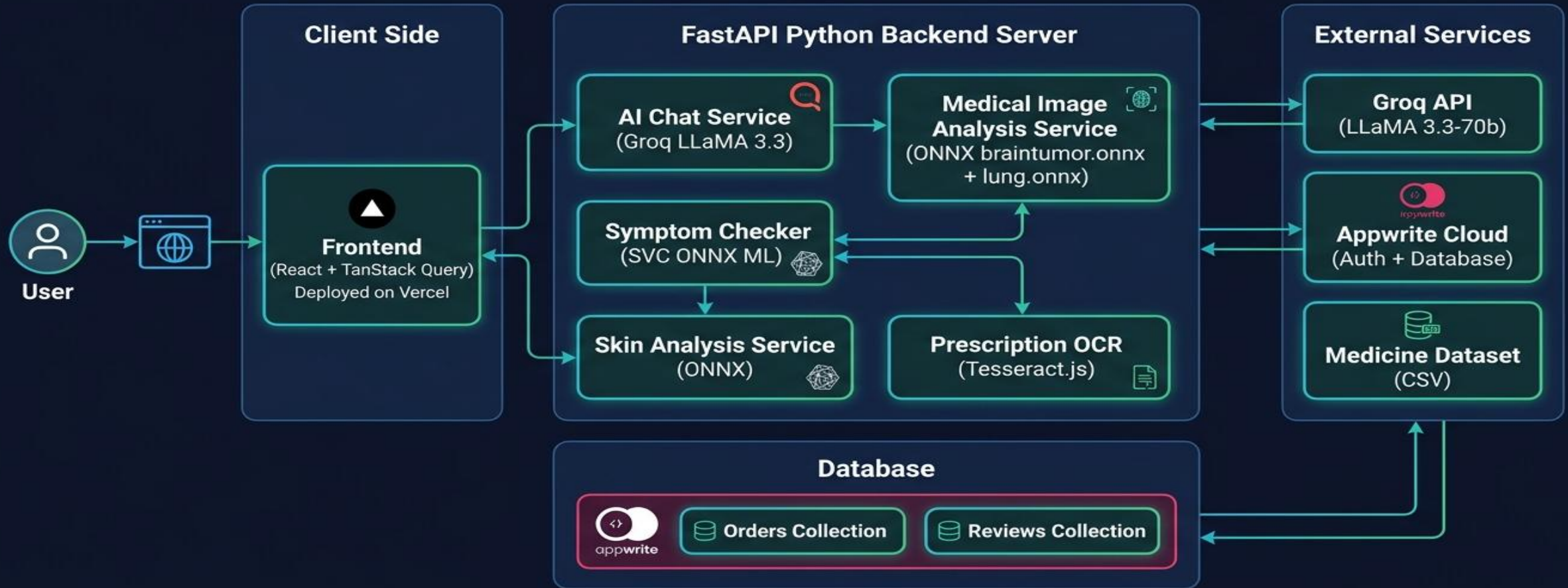
Disease-specific meal plans
Calorie & BMR calculation

Admin Dashboard

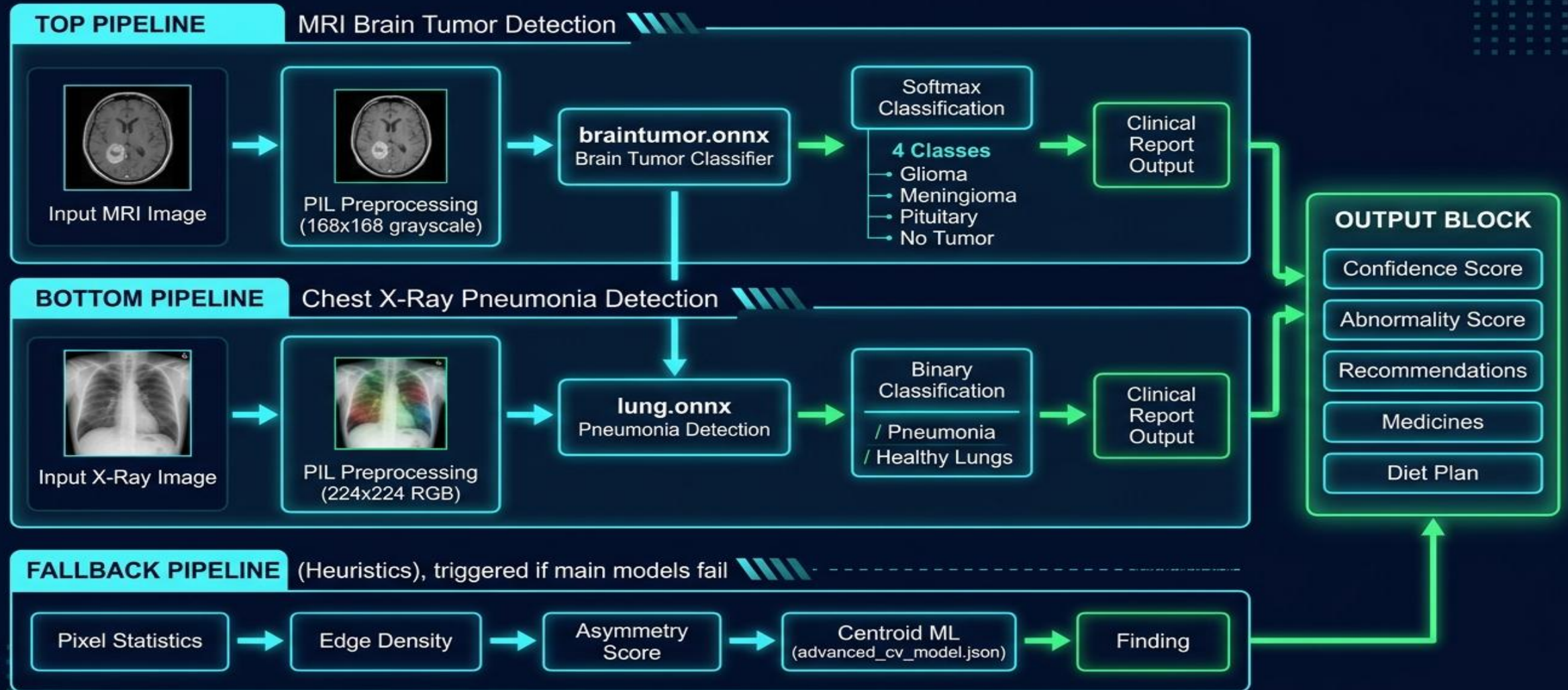
Revenue, orders, AI stats
Role-gated via Appwrite

"One platform. Every healthcare need."

SehatSaathi AI Healthcare Platform

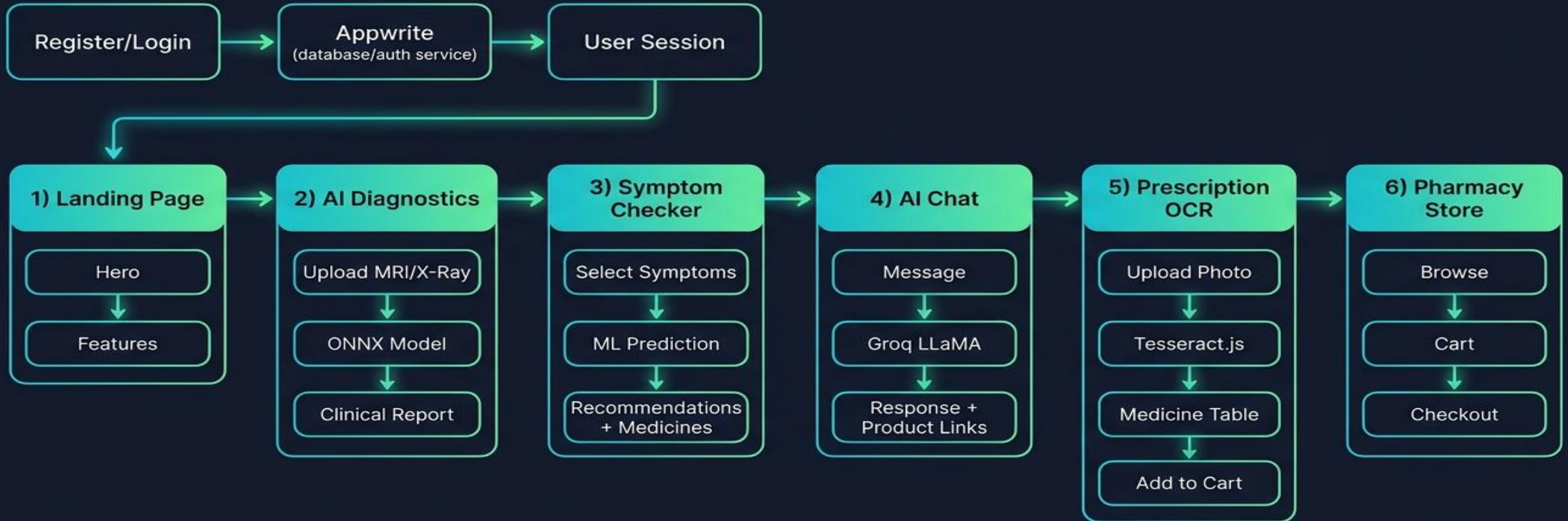


AI / ML Pipeline



I User Healthcare Application Journey

Authentication Flow



AI Chat Assistant — Bilingual (EN / اردو)

GB English Mode

User: "I have chest pain for 2 days"

SehatBot: "Chest pain lasting more than a few minutes can indicate cardiac issues. If the pain is severe or spreading to your arm, please seek emergency care immediately.

Please consult a qualified healthcare professional for accurate diagnosis."

اردو موڈ PK

"دن سے سینے میں درد ہے 2 مجھے " : صارف

سینے میں درد سنجیدہ " : سیپت ساتھی
ہو سکتا ہے۔ اگر درد شدید ہو یا
بازو تک پھیلے تو فوری طور پر
"ایمرجنسی سے رابطہ کریں۔

Powered by: Groq LLaMA 3.3-70b

Speed: ~500 tokens/second

Prescription OCR Scanner

INPUT

 Photo of handwritten or printed prescription

Drag & drop or click to upload on any device

Image compressed to max 1200px for WASM

PROCESSING

⚙ Tesseract.js WASM
(100% offline, in browser)

⚙ Image → raw text

⚙ Heuristic parser:
tab / cap / syp / inj
mg / od / bd / tds

⚙ Fallback: OCR.space API

OUTPUT

✅ Structured medicine table

Amoxicillin 500 Capsule
Paracetamol 500 Tablet
Brufen 400 Tablet

[+ Cart] per medicine

[Add ALL to Cart]

→ Checkout → COD Delivery

Technology Stack

Frontend



React 19

React navigation, forms, and components



TanStack Router

TanStack router to the browser



TanStack Start



Framer Motion

Framer Motion to the browser



Tailwind CSS Shadcn/UI

Tailwind CSS Shadcn/UI styling



TypeScript

TypeScript banners



Vite 8

Vite 8 for support

Backend & AI



FastAPI Python

Python for automations



ONNX Runtime

ONNX Runtime



Tesseract.js OCR

Tesseract.js OCR



Groq LLaMA 3.3-70b

Groq LLaMA 3.3-70b



Pillow/PIL



NumPy



ONNX Models

(braintumor, lung, skin, svc)

Infrastructure



Vercel Deployment

Vercel deployment



Appwrite Cloud

Appwrite for the cloud



Cloudflare

Cloudflare for the cloud



Node.js 24

Node.js 24 for the cloud



Nitro Server

Nitro server for the cloud



GitHub

GitHub and the cloud

Live Demo

<https://sehat-saathi-ai.vercel.app>

- STEP 1 → /diagnostics — Upload a brain MRI or chest X-Ray image
- STEP 2 → /symptoms — Select: fever, headache, nausea → predict disease
- STEP 3 → /chat — Ask a health question in Urdu
- STEP 4 → /prescription — Upload a prescription photo → OCR → Cart
- STEP 5 → /pharmacy — Browse medicines → Checkout → COD

GitHub: github.com/maryamtahir7/SehatSaathi-AI

Impact & Metrics

~99%

Brain Tumor
Model Accuracy

132

Symptoms
Supported

15,000+

Medicines
in Pharmacy

< 5 sec

Full AI
Diagnosis Time

41

Diseases
Predicted

2 langs

English
& Urdu

What We Built & What's Next

✓ DELIVERED

- ✓ 8 AI Healthcare Modules
- ✓ 4 ONNX ML Models (trained)
- ✓ Bilingual EN / Urdu (RTL)
- ✓ 15,000+ Medicine Database
- ✓ Live on Vercel — production
- ✓ Full E-Commerce Cart + COD
- ✓ Appwrite Auth & Database
- ✓ Admin Dashboard

🚀 ROADMAP

- v1.1 → Voice input (Web Speech API)
- v1.1 → WhatsApp Bot for rural users
- v1.2 → Telemedicine video calls
- v1.2 → Retinal & blood report OCR
- v2.0 → React Native mobile app
- v2.0 → Wearables integration
- v2.0 → Pakistan Govt. hospital API

"SehatSaathi AI — Making healthcare intelligent, accessible, and human."